The coordinate ring of the k-fold iterated commutator locus for 2×2 matrices[1]

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Abstract

For 2×2 matrices $A_1, ..., A_k$, write $[A_1, ..., A_k]$ for the left-normed iterated commutator $[...[[A_1, A_2], A_3], ..., A_k]$, and I_k for the ideal, in the $3k$-variable reduced-coordinate polynomial ring R_k, cutting out its vanishing locus. We prove, for every $k \geq 2$ over any field of characteristic $\neq 2$, and as four independently-established results rather than one bundled claim: I_k has codimension 2; I_k has exactly 3 minimal generators; R_k/I_k is Cohen-Macaulay; and I_k is radical. Combined with an explicit non-containment argument for the component count, these assemble into the **Primary Decomposition Theorem**: $I_k = P_2 \cap \cdots \cap P_k$ is an irredundant primary decomposition into exactly $k-1$ radical primes, following an explicit recursive block-involvement pattern. The proof identifies I_k as the ideal of 2×2 minors of an explicit 2×3 matrix (a determinantal ideal, not merely determinantal-**looking**), and invokes classical determinantal-ideal theory (Bruns-Vetter) and an explicit rank-2 Jacobian witness on every component (Serre's criterion) for the algebraic and radicality halves respectively.

1. Introduction

2×2 matrices A, B have commutator $[A, B] = AB - BA$, and the **iterated** left-normed commutator of k matrices is $[A_1, ..., A_k] := [...[[A_1, A_2], A_3], ..., A_k]$, matching the standard lower-central-series convention for Lie algebras. This object arose out of a question in algebraic complexity theory – how many scalar multiplications are needed to **compute** $[A_1, ..., A_k]$? – which remains genuinely open for $k \geq 3$ and is treated in full in the companion note *k-iterated-commutators-complexity.typ*. The one fact from that story needed here is purely algebraic, not complexity-theoretic: an explicit bilinear formula for the ordinary 2×2 commutator, chained across k matrices, gives an explicit polynomial parametrization of $[A_1, ..., A_k]$'s reduced coordinates (Definition 3 below), and it is the **vanishing locus** of this parametrization, as an algebraic variety and its coordinate ring, that is the actual subject of this article. (One naming note: the filename this article is kept under, *ring-of-k-commuting-2x2-matrices*, uses "commuting" in this generalized iterated-commutator-vanishing sense – $[A_1, ..., A_k] = 0$ – not literally "every pair of $A_1, ..., A_k$ commutes," a weaker condition than the classical pairwise commuting variety it specializes to at $k = 2$.)

The classical commuting variety. For A, B arbitrary (not necessarily 2×2) $n \times n$ matrices, the variety $C_n := \{(A, B) \in M_n \times M_n : AB = BA\}$ – the ordinary, unreduced, un-iterated case this article's own $k = 2$ specializes to – has been studied since the 1950s. Its irreducibility was proven independently by Motzkin and Taussky [3] and by Gerstenhaber [5], the latter also establishing the now-standard bound $\dim_{\mathbb{F}} \mathbb{F}[A, B] \leq n$ for the subalgebra generated by two commuting matrices. Whether the coordinate ring $\mathbb{F}[A, B]/I$, I the ideal generated by the entries of $AB - BA$, is **reduced** and **Cohen-Macaulay** for every n is a much harder question, attributed to Michael Artin and Melvin Hochster – an unpublished, informally-circulated conjecture with no single dateable original source [6], stated precisely as "Conjecture 1.1" in Majidi-Zolbanin and Snapp [7]. (A closely related, more specifically dated 1984 reducedness conjecture, for a different but adjacent ideal built from a commutator matrix's diagonal/anti-diagonal entries, is attributed to Hochster alone by Chau [8].) Freyja Hreinsdóttir's body of work [9] – building on her own 1997 Stockholm doctoral thesis – attacks the Artin-Hochster conjecture case by case at $n = 3, 4$: syzygies, Betti numbers, the canonical module, and a Koszul-dual structure theorem, verified computationally through $n = 7$ but proven in general only in these small cases. Knutson [10] studies the minimal primes of closely related schemes from a different angle. Cohen-Macaulayness of the **classical** commuting variety therefore remains open, for every $n \geq 3$, at the time of writing.

At $n = 2$ specifically – this article's $k = 2$ base case – the question is comparatively easy, and is in effect answered again, as a special case, by Theorem 31 below: I_2 (Definition 3) is literally the ideal of the $n = 2$

[1] Working draft, 2026-08-16. Uses the community "theorion" package for theorem environments – this project's standing default going forward, after a same-day side-by-side comparison against hand-rolled macros; the retired hand-rolled version is kept, not deleted, at */TRASH/fast-commutators-ring-of-k-commuting-2x2-matrices.typ* at the workspace top level. Every result cites only strictly-earlier results or external literature, no forward references. See *working-notes/proof-plan-conjectures.org*, Steps 0, 4, and 5, and *working-notes/topological-irreducibility-of-image.org*, for the fuller discussions this article condenses.

commuting variety, expressed in reduced (trace-removed) coordinates rather than the full matrix-entry coordinates the classical literature typically uses. The genuinely new object in this article is the **iterated** generalization to $k \geq 3$: not the classical commuting variety at all (see the naming note above), but a structurally analogous determinantal ideal, built by chaining the same bilinear cross-product formula that governs the $n = 2$ case. It is exactly the same **flavor** of question the Artin-Hochster conjecture asks – is this natural, geometrically meaningful ideal Cohen-Macaulay? is it reduced? – for a different, recursively-defined family of ideals, and this article answers it completely and unconditionally, for every $k \geq 2$ (Theorem 43), rather than case by case or conjecturally. In that sense the results below can be read as a fully worked structural analogue of the still-open classical question, in a setting simple enough ($n = 2$, but k unbounded) to admit a complete answer.

2. Notation

Notation used across more than one later result, in order of first appearance; symbols local to a single proof are omitted.

A typographic convention, used throughout: an **overline** denotes a **vector** – an element of $\mathbb{F}^3$ (or, for R_k itself, of the free module R_k^3) – while the **same letter, plain, with a two-index comma subscript**, denotes one of that vector's three individual scalar **components**. So $\overline{v}_i \in \mathbb{F}^3$ is block i's whole reduced-coordinate vector, while its three components are the plain scalars $v_{i,1}, v_{i,2}, v_{i,3} \in \mathbb{F}$ (equivalently $v_{i,j}$ for $j = 1, 2, 3$) – never overlined, since a single scalar entry is not itself a vector. The same convention applies to $\overline{w}_j$ versus its components $w_{j,1}, w_{j,2}, w_{j,3}$.

Symbol	Meaning	Defined in
$\overline{v}$ (also $\overline{v}_i$; components $v_{i,1}, v_{i,2}, v_{i,3}$, or $v_{i,j}$)	reduced-coordinate vector in $\mathbb{F}^3$, and its blocks	§3, Setup
D	the sign twist $D = \mathrm{diag}(-1, -1, 1)$ of Theorem 1	§3, Setup (D: Lemma 37)
$\mathbb{F}$	ambient field, characteristic $\neq 2$	§3, after Definition 3′s example
R_k	the polynomial ring $\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$	Definition 3
I_k	the defining ideal, $I_k \subset R_k$	Definition 3
$\overline{w}_j$	(a) the recursive vector in $\mathbb{F}^3$; (b) as a polynomial map $\mathbb{A}^{3j} \to \mathbb{A}^3$	(a) Definition 3; (b) Lemma 8
U_j	the Zariski-open locus where $\overline{w}_j \neq 0$	Lemma 8
π	the projection $\mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ forgetting $\overline{v}_k$	Definition 14
Φ	the map $\mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}$	Definition 14
Y_1, Y_2	the two strata of $V(I_k)$	Definition 14
P_j ($j = 2, \ldots, k$)	the primary components of I_k	Definition 24
J_k	the Jacobian of $\overline{w}_k$	Lemma 37
$[u]_\times$	skew matrix with $[u]_\times v = u \times v$	Lemma 37

3. Setup: the signed encoding and the defining ideal

Step 1: discard the scalar part. Write $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ as the sum of its scalar part $(\mathrm{tr}(A)/2)I$ and its traceless remainder $A_0 := A - (\mathrm{tr}(A)/2)I = \begin{pmatrix} d_a/2 & a_{12} \\ a_{21} & -d_a/2 \end{pmatrix}$, $d_a := a_{11} - a_{22}$. Scalar matrices are central, so they never contribute to a commutator: $[cI, B] = cIB - BcI = cB - cB = 0$ for every B and every scalar c, hence $[A, B] = [A_0, B_0]$ identically – no information relevant to **any** bracket is lost by discarding the scalar part, so from here on only the traceless remainder matters.

Step 2: encode the traceless remainder. A traceless matrix $\begin{pmatrix} x & y \\ z & -x \end{pmatrix}$ has three free entries. Rather than reading them off directly (x, y, z themselves – shown in Remark 2 below to be a strictly **worse** choice, needing two separate factors of 2), take

$$\begin{aligned}\mathrm{enc}(A) := (&a_{11} - a_{22},\\ &a_{12} + a_{21},\\ &a_{12} - a_{21}) \in \mathbb{F}^3,\end{aligned} \tag{1}$$

the traceless $(1,1)$-entry doubled (the quantity d_a of Step 1), and A's two off-diagonal entries replaced by their sum and difference. As with the naive encoding red of Remark 2, no division is needed to **compute** enc – only its inverse (recovering a_{12}, a_{21} individually as $\frac{v_2+v_3}{2}$ and $\frac{v_2-v_3}{2}$) needs 2 invertible, which is the standing hypothesis the article already carries, not a new one.

Step 3: form the commutator. Given traceless $A_0 = \left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$ and $B_0 = \left(\begin{smallmatrix} u & v \\ w & -u \end{smallmatrix}\right)$, the ordinary matrix commutator $[A_0, B_0] = A_0B_0 - B_0A_0$ is again traceless automatically ($\mathrm{tr}(PQ) = \mathrm{tr}(QP)$ for any P, Q, so $\mathrm{tr}([P,Q]) = 0$ always), hence again encodable by red.

Step 4: the encoded bracket is the cross product, up to one fixed sign twist.

Theorem 1 (the basic identity)

For traceless 2×2 matrices A_0, B_0 over $\mathbb{F}$ (char $\mathbb{F} \neq 2$),

$$\mathrm{enc}([A_0, B_0]) = D(\mathrm{enc}(A_0) \times \mathrm{enc}(B_0)), \quad D := \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{2}$$

Proof of Theorem 1. A direct computation, carried out symbolically over generic entries $a_{11}, \ldots, b_{22}$ (not merely checked at sampled points) in *code/verify_lorentzian_encoding_identity.sage*: both sides, expanded in the 8 free entries, agree term by term. □

Remark 2 (why this particular twist, and not some other one)

Three concrete respects in which D improves on the naive reshuffle $\tau(a,b,c) = (b,a,2c)$ that the encoding $\mathrm{red}(A) := (a_{12}, a_{21}, d_a)$ forces, each checked, not merely asserted:

1. ***D* does not mix coordinates.** τ swaps two of its three inputs; D acts diagonally, one coordinate at a time. Consequence: red would need a page-length remark ("why $\overline{w}_j$ carries no outer τ") to justify that rescaling+swapping a generating set by an invertible linear map doesn't change the ideal generated. Here the analogous fact is immediate: **individually** negating two of three generators obviously leaves the ideal unchanged, generator by generator – no swap to account for. (Still verified computationally, not merely asserted: $\mathrm{ideal}(\overline{w}_2) = \mathrm{ideal}(D\overline{w}_2)$, confirmed by direct Gröbner-basis equality in *code/verify_encoding_gl3_equivariance.sage*.)
2. ***D* is an involution; *τ* is not.** $D^2 = I$ exactly (immediate from the diagonal entries being ± 1), while $\tau^2 = \mathrm{diag}(1,1,4) \neq I$.
3. ***D* is not an arbitrary matrix that happens to work – it is (up to a constant) the matrix of $\mathfrak{sl}_2$'s own Killing form, in these coordinates.** For traceless $A_0 = \left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$, $\det(A_0) = -x^2 - yz$, an **indefinite** quadratic form (never definite: $-yz = -((y+z)/2)^2 + ((y-z)/2)^2$ is a genuine difference of squares whenever $2 \neq 0$). Checked directly: $-\mathrm{enc}(A)_1^2 - \mathrm{enc}(A)_2^2 + \mathrm{enc}(A)_3^2 = 4\det(A_0)$, exactly (Sage, symbolic). So D's sign pattern **is** det's own signature, read off in these coordinates – not a bookkeeping accident. (Lie-theoretically: $\mathfrak{sl}_2(\mathbb{R}) \cong \mathfrak{so}(2,1)$, the Lorentz algebra, not the compact $\mathfrak{so}(3)$ one gets only after adjoining i – classical, not cited to a specific source here, the same way this article leaves certain folklore facts uncited rather than inventing a citation.) By Sylvester's law of inertia the signature of a quadratic form is an invariant of the form itself, not of the coordinates chosen to express it – so no linear re-encoding over a field where -1 is not a square of 1 (in particular no real or rational one) can ever remove this sign entirely; D's single-sign twist is, in this precise sense, already the simplest one achievable, not merely simpler than τ by luck.

See *working-notes/alternative-encodings-musings.org* for the fuller discussion, including why the "obvious" un-permuted reading of the matrix entries is actually **worse** than either encoding (needs a swap **and** two separate factors of 2), and a remark on why the field-extension route to a literal, untwisted cross product ($i \in \mathbb{F}$, giving $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ instead) is a real alternative but a strictly narrower one.

Definition 3 (reduced iterated commutator, signed encoding)

$$\overline{w}_2 := \overline{v}_1 \times \overline{v}_2, \quad \overline{w}_j := D(\overline{w}_{j-1}) \times \overline{v}_j \quad (j = 3, \ldots, k). \tag{3}$$

Let $R_k := \mathbb{Q}[\overline{v}_1, \ldots, \overline{v}_k]$ ($\overline{v}_i \in \mathbb{Q}^3$, $3k$ variables) and $I_k \subset R_k$ the ideal generated by the three components of $\overline{w}_k$ – the vanishing ideal of $[A_1, \ldots, A_k] = 0$ in reduced coordinates. ♣

To spell out exactly what this means: "$R_k := \mathbb{Q}[\overline{v}_1, \ldots, \overline{v}_k]$" is shorthand for **the polynomial ring in the $3k$ scalar variables $v_{i,j}$ ($i = 1, \ldots, k$, $j = 1, 2, 3$)**, grouped into k blocks of 3 – **not** a ring with k vector-valued generators (a polynomial ring's generators are always scalars). At $k = 2$, written out in full:

$$R_2 = \mathbb{Q}\left[v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}\right], \tag{4}$$

six ordinary scalar indeterminates, with $\overline{v}_1 = (v_{1,1}, v_{1,2}, v_{1,3})$ and $\overline{v}_2 = (v_{2,1}, v_{2,2}, v_{2,3})$ merely a notational grouping of them into two blocks of three – convenient because $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is most naturally written as a cross product of two block-vectors, not as some formula in six unrelated scalars.

Each $\overline{w}_j$ is a triple of polynomials, i.e. $\overline{w}_j \in R_j^3$ (the free rank-3 module over R_j) – equivalently, exactly as recorded in the Notation table's entry (b) for $\overline{w}_j$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ **given by substitution**: evaluating each of $\overline{w}_j$'s three polynomial components at a point $(\overline{v}_1, \ldots, \overline{v}_j) \in \mathbb{A}^{3j}$. Concretely at $k = 2$, writing out all six variables, $\overline{w}_2$ is the substitution map

$$\begin{pmatrix} v_{1,1} \\ v_{1,2} \\ v_{1,3} \\ v_{2,1} \\ v_{2,2} \\ v_{2,3} \end{pmatrix} \in \mathbb{A}^6 \quad \mapsto \quad \begin{pmatrix} v_{1,2}v_{2,3} - v_{1,3}v_{2,2} \\ v_{1,3}v_{2,1} - v_{1,1}v_{2,3} \\ v_{1,1}v_{2,2} - v_{1,2}v_{2,1} \end{pmatrix} \in \mathbb{A}^3, \tag{5}$$

and I_k (Definition 3) is, equivalently, the ideal generated by the three coordinate functions of $\overline{w}_k : \mathbb{A}^{3k} \to \mathbb{A}^3$ pulled back along this substitution.

Example 4 ($\overline{w}_k$ and I_k, explicitly, for $k = 2, 3, 4$)

Computed directly (*code/compute_signed_encoding_examples.sage*), not hand-derived. At $k = 2$, the three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$:

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{6}$$

(Identical to the red display – the two encodings agree exactly at the very first stage, since neither τ nor D is applied yet.) At $k = 3$, the three components of $\overline{w}_3 = D(\overline{w}_2) \times \overline{v}_3$ – now with **every coefficient equal to ± 1**, no factor of 2 anywhere, unlike the τ-twisted $\overline{w}_3$ that red produces:

$$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3}, \tag{7}$$

$$-v_{1,2}v_{2,1}v_{3,1} + v_{1,1}v_{2,2}v_{3,1} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3}, \tag{8}$$

$$v_{1,3}v_{2,1}v_{3,1} - v_{1,1}v_{2,3}v_{3,1} + v_{1,3}v_{2,2}v_{3,2} - v_{1,2}v_{2,3}v_{3,2}, \tag{9}$$

and $I_3 := (w_{3,1}, w_{3,2}, w_{3,3})$. At $k = 4$, $\overline{w}_4 = D(\overline{w}_3) \times \overline{v}_4$: again every coefficient is ± 1; the first component of $\overline{w}_4$ is

$$-v_{1,3}v_{2,1}v_{3,1}v_{4,2} + v_{1,1}v_{2,3}v_{3,1}v_{4,2} - v_{1,3}v_{2,2}v_{3,2}v_{4,2} + v_{1,2}v_{2,3}v_{3,2}v_{4,2} \tag{10}$$

$$+v_{1,2}v_{2,1}v_{3,1}v_{4,3} - v_{1,1}v_{2,2}v_{3,1}v_{4,3} + v_{1,3}v_{2,2}v_{3,3}v_{4,3} - v_{1,2}v_{2,3}v_{3,3}v_{4,3}, \tag{11}$$

the other two of the same shape (omitted for space; all three components of $\overline{w}_4$, and $I_4 := (w_{4,1}, w_{4,2}, w_{4,3})$, in *code/compute_signed_encoding_examples.sage*'s output). ◇

Stated over $\mathbb{Q}$, but every result below holds over an arbitrary field $\mathbb{F}$ of characteristic $\neq 2$. Where that hypothesis is actually spent is not where one first expects, and the answer leaves a question we are happy to hand on:

Problem 5 (characteristic 2)

The hypothesis is **not** spent on D. Since $\det D = 1$, the twist is invertible over every field, and Remark 2′s "an invertible change of generators does not change the ideal" needs no hypothesis whatsoever. Under the naive reshuffle it would: $\det \tau = -2$, so τ is singular in characteristic 2 precisely.

It is spent in **Step 1**, earlier than Step 2′s division by 2 and more seriously. In characteristic 2 one has $\operatorname{tr}(cI) = 2c = 0$, so **scalar matrices are themselves traceless**. The splitting $A = (\operatorname{tr}(A)/2)I + A_0$ then has no meaning at all: the scalars do not sit beside a complement, they sit **inside** the traceless matrices.

The bracket is unbothered — scalars are central in every characteristic — so $[\cdot,\cdot]$ still descends to the three-dimensional quotient $\mathfrak{gl}_2/(\mathbb{F}\cdot I)$. The question is whether the rest of the article descends with it.

Question. Over a field of characteristic 2, is there a linear isomorphism $\mathfrak{gl}_2/(\mathbb{F}\cdot I) \cong \mathbb{F}^3$ carrying the induced bracket to a twisted cross product $E(\cdot\times\cdot)$ for some invertible E? And if so, do Theorem 19, Theorem 12, Theorem 31, Theorem 42 and Theorem 43 hold verbatim?

What makes this more than idle curiosity: **every polynomial displayed in Example 4 has coefficients ± 1 only**. No 2 appears anywhere in the output. Whatever the obstruction is, it lives entirely in the setup and not in the ideal — which is exactly the situation in which a different setup might dissolve it.

4. The determinantal structure and the generator count

Theorem 6 (determinantal structure)

For $k \geq 3$, I_k is exactly the ideal of 2×2 minors of the 2×3 matrix

$$\begin{pmatrix} D(\overline{w}_{k-1}) \\ \overline{v}_k \end{pmatrix}, \tag{12}$$

i.e. $V(I_k)$ is the preimage, under $(\overline{v}_1,\ldots,\overline{v}_k) \mapsto \begin{pmatrix} D(\overline{w}_{k-1}(\overline{v}_1,\ldots,\overline{v}_{k-1})) \\ \overline{v}_k \end{pmatrix}$, of the classical rank-$\leq 1$ determinantal variety of 2×3 matrices. (At $k=2$ the same statement holds with $D(\overline{w}_{k-1})$ replaced by $\overline{v}_1$ itself, matching Definition 3′s base case $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ directly.)

Proof of Theorem 6. Immediate from the fact that the three components of a cross product $a\times b$ are (up to sign) exactly the 2×2 minors of the 2×3 matrix with rows a, b, combined with Definition 3′s recursion – and checked independently: comparing an independent implementation of the D/cross-product recursion against the ideal actually produced by the chained bilinear formula gives the **exact same ideal** (Gröbner-basis containment both ways) at $k = 2,\ldots,6$ (*code/verify_determinantal_structure_all_k.sage*); the underlying polynomial identity $\operatorname{enc}([A,B]) = D(\operatorname{enc}(A)\times\operatorname{enc}(B))$ of Theorem 1 makes this a two-line induction valid for **every** k, not merely a pattern checked at small k (*code/verify_lorentzian_encoding_identity.sage*). □

This theorem is the engine behind the geometric results of §5: R_k/I_k is not merely **determinantal-looking**, it is literally an iterated extension of the coordinate ring of a classical determinantal variety.

Example 7 (the 2×3 matrix of Theorem 6, for $k = 2, 3, 4$)

At $k = 2$:

$$\begin{pmatrix} \overline{v}_1 \\ \overline{v}_2 \end{pmatrix} = \begin{pmatrix} v_{1,1} & v_{1,2} & v_{1,3} \\ v_{2,1} & v_{2,2} & v_{2,3} \end{pmatrix}. \tag{13}$$

At $k = 3$, the first row is $D(\overline{w}_2) = (-w_{2,1}, -w_{2,2}, w_{2,3})$ where $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is as in the previous example, giving

$$\begin{pmatrix} D(\overline{w}_2) \\ \overline{v}_3 \end{pmatrix} = \begin{pmatrix} v_{1,3}v_{2,2} - v_{1,2}v_{2,3} & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} & -v_{1,2}v_{2,1} + v_{1,1}v_{2,2} \\ v_{3,1} & v_{3,2} & v_{3,3} \end{pmatrix}. \tag{14}$$

At $k = 4$, the first row is $D(\overline{w}_3)$, entries cubic with up to 4 terms:

$$\begin{pmatrix} -v_{1,2}v_{2,1}v_{3,2} + v_{1,1}v_{2,2}v_{3,2} + v_{1,3}v_{2,1}v_{3,3} - v_{1,1}v_{2,3}v_{3,3} & \cdots & \cdots \\ v_{4,1} & v_{4,2} & v_{4,3} \end{pmatrix} \tag{15}$$

(the other two entries of the first row, of the same shape, omitted for width – all displayed in full in *code/verify_determinantal_structure_all_k.sage*). The row degree grows by one at each step $(1, 2, 3, \ldots)$, while the second row is always the fresh, free vector $\overline{v}_k$ – exactly the "one fixed complicated row, one free row" shape Proposition 29 needs. ◇

Lemma 8 (non-degeneracy)

For every $j \geq 2$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ is not identically zero. Hence $U_j := \{p \in \mathbb{A}^{3j} : \overline{w}_j(p) \neq 0\}$ is a nonempty Zariski-open subset. ♡

Proof of Lemma 8. By induction: $\overline{w}_2(e_1, e_2) = e_1 \times e_2 = e_3 \neq 0$; given $c := \overline{w}_{j-1}(p) \neq 0$, at least one standard basis vector is not proportional to $D(c)$, and a cross product vanishes only on linearly dependent pairs, so $\overline{w}_{j(p,e_i)} = D(c) \times e_i \neq 0$ for a suitable e_i. □

The generator count of I_k needs nothing beyond the definition and non-degeneracy – not codimension, not the determinantal-structure theorem above, not any external citation:

Lemma 9 (independent same-degree generators are minimal)

If a homogeneous ideal in a graded ring is generated by finitely many forms of the same degree that are linearly independent over the base field, that generating set is minimal. ♡

Proof of Lemma 9. A polynomial-coefficient combination reproducing one of the generators from the others, while staying in the same degree, can only use degree-0 (constant) coefficients – so a redundant generator would violate linear independence directly. □

Lemma 10 (the three generators of I_k are independent)

The three components of $\overline{w}_k$ are linearly independent over $\mathbb{F}$, for every $k \geq 2$. ♡

Proof of Lemma 10. By induction on k, and the induction hypothesis must be **independence** itself; see Remark 11 for why the weaker non-degeneracy of Lemma 8 does not suffice.

Base case $k = 2$. The three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ are the 2×2 minors of the generic 2×3 matrix with rows $\overline{v}_1, \overline{v}_2$, displayed in Example 4. Each contains a monomial the other two do not — $v_{1,2}v_{2,3}$ occurs only in the first, $v_{1,3}v_{2,1}$ only in the second, $v_{1,1}v_{2,2}$ only in the third — so no nontrivial constant relation is possible.

Induction step. Suppose the components of $\overline{w}_{k-1}$ are linearly independent over $\mathbb{F}$, and let $c \in \mathbb{F}^3$ satisfy $c \cdot \overline{w}_k = 0$. Writing $a := D(\overline{w}_{k-1})$, the scalar triple product identity gives

$$0 = c \cdot (a \times \overline{v}_k) = (c \times a) \cdot \overline{v}_k. \tag{16}$$

The variables $\overline{v}_k$ are a **fresh** block, occurring in neither a nor c, and the right-hand side is linear in them; so each of its three coefficients vanishes separately, i.e.

$$c \times a = 0. \tag{17}$$

Suppose $c \neq 0$. Then $c \times a = 0$ forces $a = f \cdot c$ for a single polynomial f — over the fraction field $c \times a = 0$ says a lies in the line spanned by c, and $f = a_i/c_i$ is a polynomial for any index with $c_i \neq 0$. But then a's three components are pairwise proportional, hence linearly **dependent** over $\mathbb{F}$ (any d with $d \cdot c = 0$ gives $d \cdot a = (d \cdot c)f = 0$, and such a nonzero d exists since c spans only a line in $\mathbb{F}^3$). Since D is invertible, the components of $\overline{w}_{k-1} = D^{-1}(a)$ are then linearly dependent too, contradicting the induction hypothesis. Hence $c = 0$.

Corroborated computationally: the coefficient matrix has rank 3 at $k = 2, \ldots, 6$ (*code/verify_generator_independence.sage*). □

Remark 11 (why non-degeneracy is not enough here)

It is tempting to argue instead that a constant relation would force $D(\overline{w}_{k-1})$ to be a **constant vector**, and to rule that out by Lemma 8. That argument is not valid. The relation forces only $c \times D(\overline{w}_{k-1}) = 0$, which makes the components of $D(\overline{w}_{k-1})$ pairwise proportional — not constant. A vector such as $(f, 2f, 3f)$ with $f \neq 0$ is nonvanishing, so Lemma 8 does not exclude it, yet its components are visibly dependent. Non-degeneracy is strictly weaker than what the step needs, which is why the induction above carries linear independence itself.

Theorem 12 (I_k has exactly 3 minimal generators)

For every $k \geq 2$: I_k has exactly 3 minimal generators.

Proof of Theorem 12. Immediate from Lemma 9 applied to Lemma 10's independence, since I_k is generated by the three (same-degree-k-homogeneous) components of $\overline{w}_k$ by definition. □

5. The geometric structure of $V(I_k)$

Lemma 13 (continuous images and closures preserve irreducibility)

Let $f : X \to Y$ be a continuous map of topological spaces with X irreducible. Then $f(X)$ (with the subspace topology) is irreducible, and hence so is $\overline{f(X)}$.

Proof of Lemma 13. **Image.** Suppose $f(X) = C_1 \cup C_2$ with C_1, C_2 closed in $f(X)$. By continuity, $f^{-1}(C_1)$ and $f^{-1}(C_2)$ are closed in X, and $X = f^{-1}(C_1) \cup f^{-1}(C_2)$. Since X is irreducible, one of the two equals X – say $f^{-1}(C_1) = X$ – so $f(X) \subseteq C_1$, i.e. $C_1 = f(X)$, not proper. Hence $f(X)$ is irreducible. **Closure.** If $A \subseteq Y$ is irreducible and $\overline{A} = C_1 \cup C_2$ with C_1, C_2 closed in Y, then $A = (A \cap C_1) \cup (A \cap C_2)$; irreducibility of A forces, say, $A \subseteq C_1$; since C_1 is closed, $\overline{A} \subseteq C_1$, and combined with $C_1 \subseteq \overline{A}$, $C_1 = \overline{A}$, not proper. Apply both parts with $A := f(X)$. □

This is purely topological – continuity of f is the only hypothesis used, nothing about f being a polynomial map or morphism of varieties [11] (working-notes/ topological-irreducibility-of-image.org has the fuller discussion, including why this observation matters for a possible future Lean formalization of this article).

Definition 14 (the strata Y_1, Y_2)

For $k \geq 3$, write π for the projection **forgetting $\overline{v}_k$**,

$$\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}, \quad \pi(\overline{v}_1, \ldots, \overline{v}_k) := (\overline{v}_1, \ldots, \overline{v}_{k-1}), \tag{18}$$

and Φ for the map

$$\Phi : \mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}, \quad \Phi(p, \lambda) := (p, \lambda D(\overline{w}_{k-1}(p))). \tag{19}$$

Define two subsets of $\mathbb{A}^{3k}$:

$$Y_1 := V(I_{k-1}) \times \mathbb{A}^3, \quad Y_2 := \overline{\Phi(\mathbb{A}^{3k-2})}. \tag{20}$$

Y_1 is the locus where the previous bracket already vanishes, with $\overline{v}_k$ completely free; Y_2 is the closure of the locus swept out as $\overline{v}_k$ ranges over scalar multiples of $D(\overline{w}_{k-1}(p))$, over all $p \in \mathbb{A}^{3(k-1)}$. At $k = 2$ there is no "previous bracket" to chain from, so neither Y_1 nor Φ is defined; by convention $Y_2 := V(I_2)$ directly there, matching Theorem 6's own base-case treatment.

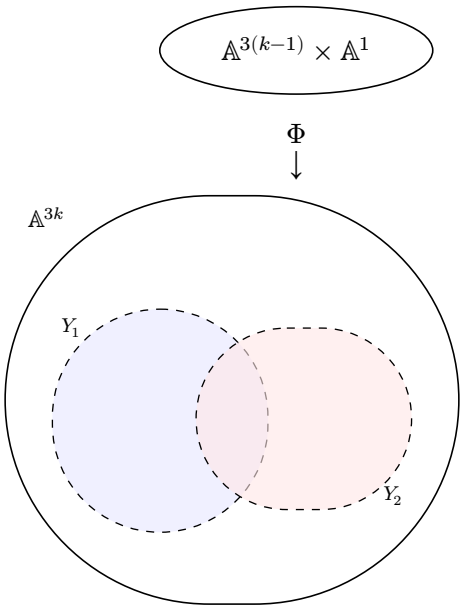


Figure 1: Both strata of $V(I_k) = Y_1 \cup Y_2$ inside the same ambient $\mathbb{A}^{3k}$: $Y_1 := V(I_{k-1}) \times \mathbb{A}^3$ (identifying $\mathbb{A}^{3(k-1)} \times \mathbb{A}^3 \cong \mathbb{A}^{3k}$) and $Y_2 := \overline{\Phi(\mathbb{A}^{3k-2})}$, Φ's image (Φ shown mapping in from above, into Y_2 specifically – Y_1 is not part of Φ's image). The two overlap (not disjoint, Propositions 17-18 rule out either containing the other): $\Phi(p, 0) = (p, 0)$ for **every** p, so the whole zero-section $\{(p, 0)\}$ lies in Y_2, and its intersection with Y_1 is $V(I_{k-1}) \times \{0\}$, of dimension $3k - 5$ – properly smaller than either stratum's own dimension $3k - 2$.

Lemma 15 (Y_2 is irreducible)

For $k \geq 3$: Y_2 (Definition 14) is irreducible, and in fact its defining ideal (the graph ideal of Φ with λ eliminated) is prime.

Proof of Lemma 15. Φ is a polynomial map, hence continuous for the Zariski topology (preimages of Zariski-closed sets under a polynomial map are again Zariski-closed, by substitution), and its domain $\mathbb{A}^{3k-2}$ is irreducible (the polynomial ring over $\mathbb{F}$ is a domain, so the zero ideal is prime). Lemma 13 gives Y_2 irreducible directly. Primality of the elimination ideal follows the same way: it is the kernel of a map into a domain ($\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]/(\cdot) \cong \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_{k-1}, \lambda]$), hence prime – this holds for every k by this argument alone, additionally corroborated computationally at $k = 4$ by computing the elimination ideal directly and confirming primality in Sage, as an independent check rather than a substitute for the argument. □

Example 16 (the defining ideals of Y_1 and Y_2, for $k = 2, 3, 4$)

Writing $\overline{v}_i = (v_{i,1}, v_{i,2}, v_{i,3})$ for the three coordinates of block $\overline{v}_i$: at $k = 2$, $Y_2 := V(I_2)$ (the base-case convention above), so its ideal **is** I_2, generated by

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{21}$$

(Y_1 is not defined at $k = 2$, Definition 14.) For $k \geq 3$, Y_1's defining ideal is simply I_{k-1} itself, viewed inside R_k by leaving $\overline{v}_k$ free – at $k = 3$, exactly the three generators displayed above for I_2, now inside R_3 with $\overline{v}_3$ not appearing at all; at $k = 4$, exactly I_3's three generators (displayed below in this same example) inside R_4 with $\overline{v}_4$ free. Unlike Y_2, no elimination is needed to compute Y_1's ideal – it is already I_{k-1}'s own generating set, unchanged. At $k = 3$, Y_2's elimination ideal (computed directly, not inferred) is generated by

$$v_{2,1}v_{3,1} + v_{2,2}v_{3,2} - v_{2,3}v_{3,3}, \quad v_{1,1}v_{3,1} + v_{1,2}v_{3,2} - v_{1,3}v_{3,3}, \tag{22}$$

$$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3} \tag{23}$$

– notably, two of the three generators are **linear** in $\overline{v}_3$ (exactly the two components of the "proportionality" condition $\overline{v}_3 \propto D(\overline{w}_2)$), and only the third, not involving $\overline{v}_3$ at all, carries the full complexity of $D(\overline{w}_2)$ itself. **This ideal is a proper factor of I_3, not equal to it** – I_3 is the intersection of this ideal with a second prime (the "Y_1" piece, where the $k = 2$ bracket already vanishes); see the component-count example after Theorem 23 below, where this exact ideal reappears as one of I_3's two primary components. At $k = 4$, by the same pattern, two of the three generators are linear in $\overline{v}_4$ and the third is not, now degree 4:

$$v_{3,1}v_{4,1} + v_{3,2}v_{4,2} - v_{3,3}v_{4,3}, \tag{24}$$

$$v_{1,3}v_{2,2}v_{4,1} - v_{1,2}v_{2,3}v_{4,1} - v_{1,3}v_{2,1}v_{4,2} + v_{1,1}v_{2,3}v_{4,2} + v_{1,2}v_{2,1}v_{4,3} - v_{1,1}v_{2,2}v_{4,3}, \tag{25}$$

$$v_{1,3}v_{2,1}v_{3,1}v_{4,2} - v_{1,1}v_{2,3}v_{3,1}v_{4,2} + v_{1,3}v_{2,2}v_{3,2}v_{4,2} - v_{1,2}v_{2,3}v_{3,2}v_{4,2} \tag{26}$$

$$-v_{1,2}v_{2,1}v_{3,1}v_{4,3} + v_{1,1}v_{2,2}v_{3,1}v_{4,3} - v_{1,3}v_{2,2}v_{3,3}v_{4,3} + v_{1,2}v_{2,3}v_{3,3}v_{4,3}. \tag{27}$$

All computed directly (elimination of λ from the graph ideal of Φ, in Sage), not just asserted from the pattern. ◇

Remark 17 (the closure, not the raw image, is what is needed)

A polynomial map's image need not itself be closed – the standard example is $\mathbb{A}^2 \to \mathbb{A}^2$, $(x, y) \mapsto (x, xy)$, whose image is not even locally closed, though it is **constructible** (a finite union of locally closed sets) by Chevalley's theorem, which has no purely topological analogue for general continuous maps [11]. This never causes a problem for Y_2, because Y_2 is **defined** as $\overline{\Phi(\cdot)}$ from the outset, and the decomposition used in Theorem 19′s proof below, $V(I_k) = Y_1 \cup Y_2$, is established by a direct case split rather than by taking closures and hoping: a point $(p, \overline{v}_k) \in V(I_k)$ either has $D(\overline{w}_{k-1}(p)) = 0$, putting it in Y_1 directly, or has $\overline{v}_k = \lambda D(\overline{w}_{k-1}(p))$ for some λ, so $(p, \overline{v}_k) = \Phi(p, \lambda)$ is **literally** a value of Φ – not merely a limit point – hence in Y_2 with no appeal to closure at all. The reverse containment $Y_1 \cup Y_2 \subseteq V(I_k)$ needs only that $\Phi(\cdot) \subseteq V(I_k)$ (a cross product of a vector with itself vanishes: $D(\overline{w}_{k-1}) \times (\lambda D(\overline{w}_{k-1})) = 0$ identically) together with $V(I_k)$ already being closed, automatic since it is a vanishing locus by definition. So the theorem never needs "the image of Φ is closed" – only that it is contained in the already-closed $V(I_k)$, which is immediate. ◇

Lemma 18 (dimension of Y_2, and where it lives)

With Y_2 as in Lemma 15 and U_{k-1} as in Lemma 8: $\dim Y_2 = 3k - 2$, and Y_2 meets $\pi^{-1}(U_{k-1})$ nonemptily, where π is the projection of Definition 14, forgetting $\overline{v}_k$. ♡

Proof of Lemma 18. $\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ discards the last block: $\pi(\overline{v}_1, \ldots, \overline{v}_k) = (\overline{v}_1, \ldots, \overline{v}_{k-1})$. Hence $\pi^{-1}(U_{k-1}) = U_{k-1} \times \mathbb{A}^3$, the set of points whose **first** $k - 1$ blocks lie in U_{k-1} and whose $\overline{v}_k$ is completely unconstrained. Over U_{k-1}, Φ is injective. Both p and λ can be recovered from $\Phi(p, \lambda)$: p directly, and λ by dividing any nonzero coordinate of $\overline{v}_k = \lambda D(\overline{w}_{k-1}(p))$ by the corresponding coordinate of $D(\overline{w}_{k-1}(p))$; this division is well-defined since $D(\overline{w}_{k-1}(p)) \neq 0$ on U_{k-1}. Injectivity gives $\dim Y_2 = \dim(U_{k-1} \times \mathbb{A}^1) = 3(k-1) + 1 = 3k - 2$. Since U_{k-1} is nonempty by Lemma 8, $\Phi(U_{k-1} \times \{1\})$ is a nonempty subset of $Y_2 \cap \pi^{-1}(U_{k-1})$. □

Theorem 19 (I_k has codimension 2)

For every $k \geq 2$: $\dim V(I_k) = 3k - 2$, i.e. $\operatorname{codim}(I_k) = 2$.

Proof of Theorem 19. Induction on k. **Base case $k = 2$**: $V(I_2)$ is the rank-≤ 1 locus of a 2×3 matrix with independent-indeterminate entries, of the standard dimension $4 = 3 \cdot 2 - 2$ for the generic determinantal variety of this shape [3]. **Inductive step**: by Theorem 6, and since $a \times b = 0$ if and only if a, b are linearly dependent, $V(I_k) = Y_1 \cup Y_2$ (Definition 14). The locus $D(\overline{w}_{k-1}) = 0$ gives Y_1, with $\overline{v}_k$ unconstrained; the locus $D(\overline{w}_{k-1}) \neq 0$ with $\overline{v}_k$ proportional to it gives a subset of Y_2. This is the direct case split of Remark 17, not merely a closure-level approximation. By the inductive hypothesis, $\dim V(I_{k-1}) = 3(k-1) - 2$, so $\dim Y_1 = 3(k-1) - 2 + 3 = 3k - 2$. Also $\dim Y_2 = 3k - 2$, by Lemma 18. Hence $\dim V(I_k) = \dim(Y_1 \cup Y_2) = 3k - 2$. This dimension statement alone needs no containment argument; that is needed only for the sharper component-count result below. □

Example 20 (dimension and codimension, for $k = 2, 3, 4$)

Values of $\dim R_k$ (the ambient dimension, $= 3k$), $\dim(R_k/I_k)$ (the dimension of the variety $V(I_k)$ – **not** "$\dim I_k$", an ideal is not a variety and does not itself have a dimension in this sense), and $\operatorname{codim}(I_k) := \dim R_k - \dim(R_k/I_k)$:

k	$\dim R_k$	$\dim(R_k/I_k)$	$\operatorname{codim}(I_k)$
2	6	4	2
3	9	7	2
4	12	10	2

All three rows follow directly from Theorem 19's formula $\dim(R_k/I_k) = 3k - 2$; also confirmed by direct Gröbner-basis dimension computation in Sage at each k, as an independent check.

Proposition 21 ($Y_2 \not\subseteq Y_1$)

With Y_1, Y_2 as in the proof of Theorem 19: $Y_2 \not\subseteq Y_1$.

Proof of Proposition 21. Pick $p \in U_{k-1}$ (Lemma 8); then $\Phi(p, 0) = (p, 0) \in Y_2$ but $\overline{w}_{k-1}(p) \neq 0$ puts $(p, 0) \notin Y_1$. □

Proposition 22 (no component of Y_1 lies in Y_2)

No irreducible component Z of $V(I_{k-1})$ (each of dimension $3(k-1) - 2$, by Theorem 19 applied at $k - 1$) has $Z \times \mathbb{A}^3 \subseteq Y_2$.

Proof of Proposition 22. Both $Z \times \mathbb{A}^3$ and Y_2 are irreducible: Y_2 by Lemma 15, and $Z \times \mathbb{A}^3$ because a product of irreducible varieties is irreducible (Z is irreducible by hypothesis, and $\mathbb{A}^3$ trivially is). Both have the **same** dimension $3k - 2$, by Theorem 19 and Lemma 18. Containment would therefore force equality. This is refuted: $Z \times \mathbb{A}^3$ is disjoint from $\pi^{-1}(U_{k-1})$, since $\overline{w}_{k-1}$ vanishes identically on $Z \subset V(I_{k-1})$; but Y_2 meets $\pi^{-1}(U_{k-1})$, by Lemma 18. □

Theorem 23 (component count)

For every $k \geq 2$: $V(I_k)$ has exactly $k - 1$ irreducible components, each of dimension $3k - 2$. The component indexed $j = 2, \ldots, k$ involves exactly the first j coordinate blocks and nothing later.

Proof of Theorem 23. Induction on k. **Base case $k = 2$**: the rank-≤ 1 locus of a generic 2×3 matrix is irreducible – the affine cone over the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$, equivalently the Motzkin-Taussky [3] irreducibility of the 2×2 commuting variety – so exactly 1 component, matching $k - 1 = 1$. **Step**: $V(I_k) = Y_1 \cup Y_2$ as in Theorem 19's proof; by the inductive hypothesis Y_1 has $k - 2$ irreducible components, each of dimension $3k - 2$ (Theorem 19), none contained in another; Proposition 21 and Proposition 22 rule out containment between Y_2 and any of them or vice versa. A finite union of irreducible closed sets with no containments is irredundant, giving exactly $(k - 2) + 1 = k - 1$ components. □

This gives I_k's minimal primes and their count, but not yet that I_k is radical (so not yet that these are literally the components of a **primary** decomposition of I_k itself, as opposed to of $\sqrt{I_k}$) – that is closed in §6, after Cohen-Macaulayness is available.

Definition 24 (the components P_j)

For $k \geq 2$ and $j = 2, \ldots, k$, write P_j for the vanishing ideal of the component of $V(I_k)$ indexed j in Theorem 23 – the one involving exactly the first j coordinate blocks. Explicitly, recursively in j:

- **Base case, $j = k$**: P_k is Y_2's own defining ideal at parameter k (Lemma 15).
- **Recursive case, $j < k$**: P_j is that same ideal computed at parameter j instead – i.e. Y_2's defining ideal when the construction is run with $k = j$ – viewed inside R_k by leaving $\overline{v}_{j+1}, \ldots, \overline{v}_k$ free.
- **Consequence**: P_j does not depend on which $k \geq j$ it is viewed inside of – it is the same ideal, just embedded in a ring with more or fewer free variables.

♣

Example 25 (a suitable point of Y_2, concretely, at $k = 3$)

Rather than only the defining ideal (Example 24 below), a single explicit point: take $\overline{v}_1 = (1, 0, 0)$, $\overline{v}_2 = (0, 1, 0)$ (so $\overline{v}_1 \in U_2$, Lemma 8), giving $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2 = (0, 0, 1)$ and $D(\overline{w}_2) = (0, 0, 1)$. Taking $\lambda = 1$ in Definition 14's Φ, i.e. $\overline{v}_3 := D(\overline{w}_2) = (0, 0, 1)$, gives the point

$$(\overline{v}_1, \overline{v}_2, \overline{v}_3) = ((1, 0, 0), (0, 1, 0), (0, 0, 1)) \in Y_2 \subset \mathbb{A}^9. \tag{28}$$

Directly: $\overline{w}_3 = D(\overline{w}_2) \times \overline{v}_3 = (0, 0, 2) \times (0, 0, 2) = 0$ (a vector crossed with itself), confirming this point lies in $V(I_3)$, as any point of the form $\Phi(p, \lambda)$ must (Definition 14). This is also, concretely, a point of the largest-dimension inherited case: it is **not** a rank-2 witness in Lemma 40's sense (that witness uses $\lambda = 0$, not $\lambda = 1$) – it illustrates Y_2 itself, not the radicality argument's own witness points.

♢

Proposition 26 (rational parametrization of each component)

For $k \geq 2$ and $j = 2, \ldots, k$, the component of $V(I_k)$ indexed j (Theorem 23) is a **rational** variety: it is the closure of the image of the polynomial map

$$\mathbb{A}^{3(j-1)} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)} \to \mathbb{A}^{3k}, \quad (p, \lambda, q) \mapsto \big(\Phi_j(p, \lambda), q\big), \tag{29}$$

where Φ_j is Φ (Definition 14) run at parameter j in Definition 24′ sense (at $j = 2$, $\Phi_2(p, \lambda) := (p, \lambda p)$, matching Theorem 6′s own base-case convention), and q fills the remaining free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$. This map is moreover **birational** onto the component: it is injective over $U_{j-1} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)}$, by the same argument as Lemma 18, applied at parameter j.

♠

Proof of Proposition 26. By Definition 24, P_j's defining ideal is Y_2's own defining ideal run at parameter j – i.e. the component itself **is** Y_2 at parameter j, crossed with the free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$. Lemma 18's injectivity argument (over U_{j-1}, via Lemma 8 at level $j - 1$) applies verbatim at parameter j, and appending the free blocks q unchanged does not affect injectivity. □

Remark 27 (a reducible variety is not itself "rational" – component-by-component is what Proposition 26 answers)

$Y_1 := V(I_{k-1}) \times \mathbb{A}^3$ (Definition 14) is **not** irreducible for $k \geq 4$ – it is a union of $k - 2$ of the components above (Theorem 23's inductive step) – so a single rational parametrization **of Y_1 as one object** is not meaningful in the usual sense (rationality is a property of irreducible varieties). What **is** meaningful, and what Proposition 26 supplies, is a rational parametrization of each of Y_1's irreducible pieces individually – exactly the components inherited from $V(I_{k-1})$.

♢

Example 28 (the components, worked out at $k = 3$)

Uniformly by k, since the components' **count** and **block pattern** are already known (Theorem 23, Definition 24) and only their explicit generators change:

- **At $k = 2$**: I_2 is already prime (Theorem 23's base case), so $V(I_2)$ has exactly $k - 1 = 1$ component – itself.
- **At $k = 3$**: exactly $k - 1 = 2$ components, both computed directly (Gröbner bases, not inferred from the pattern alone), matching $I_3 = P_2 \cap P_3$:
  - P_2, the "Y_1" piece (block pattern $\{1, 2\}$, matching that the length-2 sub-bracket already vanishes with $\overline{v}_3$ free) – generated by exactly I_2's three generators from the first example, now inside R_3 with $\overline{v}_3$ simply not appearing:
  $$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{30}$$
  - P_3, the "Y_2" piece (block pattern $\{1, 2, 3\}$) – exactly the ideal already computed in the Y_2-defining-ideal example above:
  $$v_{2,1}v_{3,1} + v_{2,2}v_{3,2} - v_{2,3}v_{3,3}, \quad v_{1,1}v_{3,1} + v_{1,2}v_{3,2} - v_{1,3}v_{3,3}, \tag{31}$$
  $$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3}. \tag{32}$$
- **At $k = 4$**: exactly $k - 1 = 3$ components, block patterns $\{1, 2\}$, $\{1, 2, 3\}$, $\{1, 2, 3, 4\}$:
  - The first two, P_2 and P_3, are inherited **verbatim** from $k = 3$ above, with $\overline{v}_4$ simply left free – not recomputed, per Definition 24′ own "does not depend on which k" clause.
  - The third, P_4, is the new "Y_2" stratum – exactly the $k = 4$ row of the earlier Y_2-defining-ideal example (generators omitted here to avoid repeating that display; same growth pattern, quartic with up to 8 terms per generator).

6. Cohen-Macaulayness

Proposition 29 (Eagon-Northcott / Hilbert-Burch, applied)

Let M be a 2×3 matrix with entries in a polynomial ring R over a field (arbitrary entries, not necessarily independent indeterminates), and J the ideal of its 2×2 minors. If J has the **expected** codimension $3 - 2 + 1 = 2$, then J is a perfect ideal of grade 2 – $\mathrm{pd}(R/J) = 2$, hence R/J is Cohen-Macaulay.

Proof of Proposition 29. Bruns and Vetter [1], Theorem (2.7), p. 13, gives exactly this for an **arbitrary** matrix, as needed since one row of our 2×3 matrix $(D(\overline{w}_{k-1}))$ is a fixed polynomial vector, not a row of independent indeterminates. (The original construction, Eagon and Northcott [2], is superseded here by Bruns-Vetter's more general treatment – confirmed closed-access via Unpaywall, not used directly.) □

Remark 30 (why the generator count above wasn't proven this way)

It is tempting to expect Proposition 29 to also hand over the generator count for free via Bruns-Vetter's Theorem (16.36), p. 217 ("The Theorem of Hilbert-Burch") – but that theorem, as actually stated, gives only $I = a \cdot I_m(f)$ for a nonzerodivisor a, not a generator count on its own: at expected codimension, an **arbitrary** 2×3 matrix's minors can be perfect and Cohen-Macaulay while one minor vanishes identically or is a scalar combination of the others (concretely: $\begin{pmatrix} x & y & 0 \\ 0 & 0 & 1 \end{pmatrix}$ over $\mathbb{F}[x, y]$ has minors $0, x, y$, height 2 as expected, $J = (x, y)$ perfect and Cohen-Macaulay with only **2** generators). This is exactly why Theorem 12 was proven directly from linear independence instead of cited from this section's machinery.

Theorem 31 (R_k/I_k is Cohen-Macaulay)

For every $k \geq 2$: R_k/I_k is Cohen-Macaulay.

Proof of Theorem 31. By Theorem 19, I_k has codimension 2, the expected value for the 2×3 matrix of Theorem 6 presenting it. Proposition 29 applies directly. (This uses Theorem 19 only – not Theorem 12's generator count, not Theorem 23's component count – genuinely independent of both.) □

7. Radicality

Cohen-Macaulayness and the explicit component structure of §5 combine, via Serre's criterion, to give radicality – which then closes both the generator-count (§4) / Cohen-Macaulay (§6) story (making I_k itself, not just $\sqrt{I_k}$, the object with those properties) and upgrades §5′s component count to a genuine primary decomposition.

Lemma 32 (reduction via Serre)

Recall Serre's condition S_n: a Noetherian ring A satisfies S_n if $\operatorname{depth}(A_P) \geq \min(n, \dim A_P)$ for every prime $P \subset A$ (Cohen-Macaulay means $\operatorname{depth}(A_P) = \dim A_P$ at **every** prime – the extremal case, $n = \infty$ in effect). Cohen-Macaulay rings satisfy S_n for every n: $\operatorname{depth}(A_P) = \dim A_P \geq \min(n, \dim A_P)$ trivially, whatever n is – this needs nothing beyond the two definitions just given, standard for Serre's conditions in general (see e.g. Bruns and Vetter [1] or Eisenbud's *Commutative Algebra with a View Toward Algebraic Geometry*). In particular S_1. Serre's criterion for reducedness – a Noetherian ring is reduced if and only if it satisfies R_0 and S_1 [12] – then says that, given Theorem 31, radicality of I_k is equivalent to R_0 alone: regularity of R_k/I_k at each of its minimal primes.

The three lemmas that follow are the standard Jacobian criterion, split into its two genuinely separate ingredients – one linear-algebraic (Lemma 33), one dimension-theoretic (Lemma 34) – before either is used. The order is deliberate: the rank bound and the regularity criterion each draw on **both** ingredients and on neither each other, which a single pair of mutually-citing lemmas (as in an earlier draft of this section) obscures.

Lemma 33 (the cotangent identity)

Let $S = \mathbb{F}[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$, $A = S/I$, and $x \in V(I)$ an $\mathbb{F}$-rational point, with $\mathfrak{n} \subset S$ the maximal ideal at x and $\mathfrak{m} = \mathfrak{n}/I \subset A$ its image. Then

$$\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - \operatorname{rank} \operatorname{Jac}(x). \tag{33}$$

Proof of Lemma 33. Linear algebra only; no hypothesis on A is needed. Sending g to its gradient at x identifies the cotangent space $\mathfrak{n}/\mathfrak{n}^2$ with $\mathbb{F}^n$. Under that identification the image of I is the span of the linear parts at x of $f_1, \ldots, f_m$: a general element of I is $\sum_i g_i f_i$, whose linear part at x is $\sum_i g_i(x)$ times that of f_i, since every f_i vanishes at x. That span is the row space of $\operatorname{Jac}(x)$, of dimension $\operatorname{rank} \operatorname{Jac}(x)$. As $\mathfrak{m}/\mathfrak{m}^2 \cong (\mathfrak{n}/\mathfrak{n}^2)/\operatorname{im}(I)$, the claim follows. □

Lemma 34 (the local dimension at a rational point)

Let $S = \mathbb{F}[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$, $A = S/I$ equidimensional of dimension $n - c$, and $x \in V(I)$ an $\mathbb{F}$-rational point with maximal ideal $\mathfrak{m} \subset A$ and Jacobian $\operatorname{Jac}(x) \in \mathbb{F}^{m \times n}$. Then $\dim A_{\mathfrak{m}} = n - c$.

Proof of Lemma 34. $\dim A_{\mathfrak{m}} = \max\{\dim (A/P)_{\mathfrak{m}} : P \in \operatorname{Min}(A), P \subseteq \mathfrak{m}\}$, and each $A/P = S/\tilde{P}$ is an affine domain over a field, for which $\dim \left(S/\tilde{P}\right)_{\mathfrak{M}} = \dim S/\tilde{P}$ at **every** maximal ideal $\mathfrak{M}$ – affine domains over a field are catenary and equicodimensional (Eisenbud, *Commutative Algebra*, Cor. 13.4; equivalently the Stacks Project, tag 00OS [4] – a dimension formula for finitely generated field extensions, holding over an arbitrary field, not needing a perfect-field hypothesis by accident). By equidimensionality every such $\dim S/\tilde{P}$ equals $n - c$. □

Lemma 35 (Jacobian criterion: the rank bound)

Hypotheses as in Lemma 34. Then $\operatorname{rank} \operatorname{Jac}(x) \leq c$.

Proof of Lemma 35. $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 \geq \dim A_{\mathfrak{m}}$ in any Noetherian local ring. By Lemma 33 the left-hand side is $n - \operatorname{rank} \operatorname{Jac}(x)$, and by Lemma 34 the right-hand side is $n - c$; rearranging gives $\operatorname{rank} \operatorname{Jac}(x) \leq c$. □

Lemma 36 (Jacobian criterion: regularity at rank $= c$)

Hypotheses as in Lemma 34. If $\operatorname{rank} \operatorname{Jac}(x) = c$, then $A_{\mathfrak{m}}$ is regular of dimension $n - c$; in particular x lies on a **unique** component of $V(I)$, whose local ring there is a field.

Proof of Lemma 36. If rank $\mathrm{Jac}(x) = c$ then $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - c$ by Lemma 33, while $\dim A_{\mathfrak{m}} = n - c$ by Lemma 34; equality of the two is the definition of regularity. A regular local ring is a domain, so $\mathfrak{m}$ contains a unique minimal prime of A, and the local ring at that prime is a further localization of a regular local ring at its own maximal ideal in dimension 0, hence a field. □

Lemma 37 (the Jacobian's own recursion)

Write J_k for the $3 \times 3k$ Jacobian of $\overline{w}_k$, $[u]_\times$ for the skew matrix with $[u]_\times v = u \times v$. Then $J_2 = \left[-[\overline{v}_2]_\times \;\middle|\; [\overline{v}_1]_\times\right]$, and for $k \geq 3$,

$$J_k = \left[-[\overline{v}_k]_\times DJ_{k-1} \;\middle|\; [D(\overline{w}_{k-1})]_\times\right]. \tag{34}$$

Proof of Lemma 37. For the second block, $\overline{w}_k = D(\overline{w}_{k-1}) \times \overline{v}_k = [D(\overline{w}_{k-1})]_\times \overline{v}_k$ is linear in $\overline{v}_k$ with coefficient matrix free of $\overline{v}_k$. For the first, differentiating $\overline{w}_k = D(\overline{w}_{k-1}) \times \overline{v}_k$ with respect to any variable t among $\overline{v}_1, \ldots, \overline{v}_{k-1}$ gives $\partial_t \overline{w}_k = (D\partial_t \overline{w}_{k-1}) \times \overline{v}_k = -[\overline{v}_k]_\times D\partial_t \overline{w}_{k-1}$, using $a \times b = -[b]_\times a$; assembling columns gives the stated product. □

Example 38 (J_3, explicitly)

With variables ordered $v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}, v_{3,1}, v_{3,2}, v_{3,3}$, the 3×9 Jacobian of $\overline{w}_3$ (the three generators of I_3 from the earlier example), column-blocked to match Lemma 37's formula $J_3 = \left[-[\overline{v}_3]_\times DJ_2 \;\middle|\; [D(\overline{w}_2)]_\times\right]$ – shown here as the two blocks separately, since side by side the full 3×9 matrix does not fit the page width. First, $-[\overline{v}_3]_\times DJ_2$ (3×6):

$$\begin{pmatrix} -v_{2,2}v_{3,2} + v_{2,3}v_{3,3} & v_{2,1}v_{3,2} & -v_{2,1}v_{3,3} & v_{1,2}v_{3,2} - v_{1,3}v_{3,3} & -v_{1,1}v_{3,2} & v_{1,1}v_{3,3} \\ v_{2,2}v_{3,1} & -v_{2,1}v_{3,1} + v_{2,3}v_{3,3} & -v_{2,2}v_{3,3} & -v_{1,2}v_{3,1} & v_{1,1}v_{3,1} - v_{1,3}v_{3,3} & v_{1,2}v_{3,3} \\ -v_{2,3}v_{3,1} & -v_{2,3}v_{3,2} & v_{2,1}v_{3,1} + v_{2,2}v_{3,2} & v_{1,3}v_{3,1} & v_{1,3}v_{3,2} & -v_{1,1}v_{3,1} - v_{1,2}v_{3,2} \end{pmatrix} \tag{35}$$

and then $[D(\overline{w}_2)]_\times$ (3×3, the skew matrix of $D(\overline{w}_2)$'s three coordinates from the earlier matrix example):

$$\begin{pmatrix} 0 & v_{1,2}v_{2,1} - v_{1,1}v_{2,2} & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} \\ -v_{1,2}v_{2,1} + v_{1,1}v_{2,2} & 0 & -v_{1,3}v_{2,2} + v_{1,2}v_{2,3} \\ v_{1,3}v_{2,1} - v_{1,1}v_{2,3} & v_{1,3}v_{2,2} - v_{1,2}v_{2,3} & 0 \end{pmatrix} \tag{36}$$

Both blocks computed directly, not merely asserted from the recursive formula.

Remark 39 (a false shortcut, caught and corrected)

A naive reading of Lemma 37 suggests appending $\overline{v}_k = 0$ preserves rank on an inherited component – this is false: the old-variable block is left-multiplied by $-[\overline{v}_k]_\times D$, not merely padded with zero rows. At the first inherited case ($k = 2 \to 3$), appending $\overline{v}_3 = 0$ actually drops the rank to 0. The correct witness (Lemma 41 below) instead picks $\overline{v}_k = e_i$ for a standard basis vector **not** lying in $D \cdot \mathrm{im}\ J_{k-1}(q)$ – a genuine choice, not a formality.

Lemma 40 (rank-2 witness on the new component)

For every $k \geq 3$, there is a point $x = (p, 0)$, p the Lemma 8 non-degeneracy witness at level $k-1$, with $x \in V(I_k)$ and rank $J_{k(x)} = 2$. At $k = 2$ the same conclusion holds with $x = (p, 0)$ for any $p \neq 0$.

Proof of Lemma 40. **The case $k = 2$ separately**, because the statement above cannot be read at $k = 2$: it would refer to a non-degeneracy witness "at level 1", and neither $\overline{w}_1$ nor Lemma 8 exists there — Definition 3 starts the recursion at $j = 2$. Nor does Lemma 37's $k \geq 3$ formula apply. Argue directly instead: $J_2 = \left[-[\overline{v}_2]_\times \;\middle|\; [\overline{v}_1]_\times\right]$ carries no D at all, so for any $p \neq 0$, $J_2(p, 0) = [0 \mid [p]_\times]$, which has rank 2 by the kernel computation below. And $(p, 0) \in V(I_2)$ since $\overline{w}_2(p, 0) = p \times 0 = 0$.

For $k \geq 3$. By Lemma 37, $J_{k(p,0)} = \left[0 \;\middle|\; [D(\overline{w}_{k-1}(p))]_\times\right]$ since $-[0]_\times DJ_{k-1}(p) = 0$; and $D(\overline{w}_{k-1}(p)) \neq 0$ since $\overline{w}_{k-1}(p) \neq 0$ (Lemma 8) and D is invertible. For $u \neq 0$, $\ker [u]_\times = \{v : u \times v = 0\} = \mathbb{F}u$ (a cross product vanishes exactly on linearly dependent pairs), so $[D(\overline{w}_{k-1}(p))]_\times$ has rank 2, giving rank $J_{k(p,0)} = 2$. □

Lemma 41 (rank-2 witness on an inherited component)

For every $k \geq 3$ and every component Z of $V(I_{k-1})$ with a rank-2 witness $q \in Z$ (Lemma 40, or this lemma applied at $k-1$), there is $x = (q, e_i) \in Z \times \mathbb{A}^3 \subset V(I_k)$ with rank $J_{k(x)} = 2$.

Proof of Lemma 41. Let $W := D \cdot \operatorname{im} J_{k-1}(q)$, a 2-dimensional subspace of $\mathbb{F}^3$ since D is invertible and rank $J_{k-1}(q) = 2$. Since e_1, e_2, e_3 span $\mathbb{F}^3$ and $W \neq \mathbb{F}^3$, some $e_i \notin W$; fix such an i. Since $q \in V(I_{k-1})$, $\overline{w}_{k-1}(q) = 0$, so Lemma 37 gives $J_{k(q,e_i)} = \left[-[e_i]_\times DJ_{k-1}(q) \;\middle|\; 0\right]$, of rank $\dim\left([e_i]_\times W\right)$ since $DJ_{k-1}(q)$ has image exactly W. As $\ker [e_i]_\times = \mathbb{F}e_i$ and $W \cap \mathbb{F}e_i = 0$ (else $e_i \in W$), $[e_i]_\times$ is injective on W, so $\dim\left([e_i]_\times W\right) = 2$. □

Theorem 42 (I_k is radical)

For every $k \geq 2$: I_k is radical.

Proof of Theorem 42. By induction on k (mirroring Theorem 23′s own induction), Lemma 40 and Lemma 41 exhibit a rank-2-Jacobian point on **every** one of the $k-1$ components of Theorem 23. Lemma 36 then gives R_0 at every one (and, by its "unique component" clause, confirms each witness lies only on its intended component – an independent check on Theorem 23′s count, not merely assumed); Lemma 32 gives S_1, using Theorem 31′s Cohen-Macaulayness. Serre's criterion [12] gives radicality. □

Verified independently in Sage, for the signed encoding, at every $k = 2, \ldots, 6$: the Jacobian recursion (Lemma 37) checked as a polynomial identity, not just at points, and every witness confirmed to lie on $V(I_k)$ with rank $J_k = 2$ (*code/referee_radicality_D_check.sage*). The sharper claim — that each witness lies on **exactly** its intended component — is confirmed by an independently-computed primary decomposition at every $k = 3, 4, 5, 6$, which returns $k-1$ components in each case, in agreement with Theorem 23; the $k = 6$ decomposition takes some hours.

8. The Primary Decomposition Theorem

Every ingredient is now in hand for the fully general statement – component count (§5) and radicality (§7) together, neither sufficient alone:

Theorem 43 (Primary Decomposition Theorem)

For every $k \geq 2$, over any field of characteristic $\neq 2$: I_k has exactly $k-1$ irreducible, radical, primary components, namely $P_2, \ldots, P_k$ of Definition 24, and

$$I_k = P_2 \cap \cdots \cap P_k \tag{37}$$

is an irredundant primary decomposition into $k-1$ primes.

Proof of Theorem 43. Spelled out in full, combining the two independent halves proven above rather than left as a one-line conjunction. By Theorem 42, I_k is radical, i.e. $I_k = \sqrt{I_k}$. A standard fact about radical ideals in a Noetherian ring: a radical ideal equals the intersection of its own minimal primes, $I_k = \cap_{P \in \operatorname{Min}(I_k)} P$ – since $\sqrt{I_k}$ is, by definition, the intersection of **all** primes containing I_k, and the minimal ones already suffice (every non-minimal prime containing I_k contains some minimal one, hence is redundant in the intersection). Theorem 23 identifies $\operatorname{Min}(I_k)$ explicitly: exactly $k-1$ primes, one per block pattern $j = 2, \ldots, k$, named $P_2, \ldots, P_k$ by Definition 24. Substituting gives the displayed equation. Each P_j, being prime, is trivially its own primary-decomposition component (P_j-primary with associated prime P_j itself) – no separate argument for "primary" beyond "prime" is needed, and primality of each P_j is exactly Lemma 15 (applied at the appropriate parameter, via Theorem 42′s own induction). Irredundancy – no P_j can be omitted, i.e. no P_j contains $\cap_{i \neq j} P_i$ – is exactly Theorem 23′s own no-containment conclusion (Proposition 17 and Proposition 18, applied at every level of that theorem's induction), not a fact needing any further proof here. □

Remark 44 (summary)

Theorem 19 (codimension 2), Theorem 12 (3 minimal generators), Theorem 31 (Cohen-Macaulay), and Theorem 42 (radical) together say exactly what a single bundled "I_k has codimension 2, is radical, has exactly 3 minimal generators, and is Cohen-Macaulay" statement would – now as four independently-proven results, each established at the point in this article where its actual dependencies (and no others) were available, rather than one claim proven in parts across several sections. ◇

9. Summary

For every $k \geq 2$, over any field $\mathbb{F}$ of characteristic $\neq 2$:

- I_k has codimension 2 (Theorem 19), presented as the ideal of 2×2 minors of an explicit 2×3 matrix (Theorem 6) – a genuine determinantal ideal, not merely determinantal-looking.
- I_k has exactly 3 minimal generators (Theorem 12), proven directly from linear independence, independent of the determinantal presentation.
- R_k/I_k is Cohen-Macaulay (Theorem 31), from the determinantal presentation and classical Eagon-Northcott/Hilbert-Burch theory (Proposition 29).
- I_k is radical (Theorem 42), via Serre's criterion: Cohen-Macaulayness gives S_1 (Lemma 32), and an explicit rank-2 Jacobian witness on every component gives R_0 (Lemma 40, Lemma 41).
- $V(I_k)$ has exactly $k-1$ irreducible components (Theorem 23), each rational – in fact birational to an explicit affine parameter space (Proposition 26) – and these assemble, via the two facts above, into the irredundant primary decomposition of the Primary Decomposition Theorem (Theorem 43): $I_k = P_2 \cap \cdots \cap P_k$.

Nothing above is conjectural or checked only at small k: every statement is proven for **every** $k \geq 2$, over an **arbitrary** field of the stated characteristic. The one running computational thread – Sage verification of the determinantal identity, the Jacobian recursion, and the witness points, at $k = 2, \ldots, 6$ – corroborates the general proofs; it is not a substitute for them anywhere in this article.

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